AgriFleet Analytics: traceable fleet maintenance on a farm

Private portfolio project. The four machines and their telemetry are simulated. The as-is process and the stakeholder needs come from the author's own farm experience and have not been validated in interviews with other farmers or workshops.

PROBLEM

On a small hay and dairy farm, machines get serviced when someone remembers a due interval or when they break during work. Nothing is recorded, and the workshop gets only a verbal description.

APPROACH

Requirements engineering end to end: stakeholder analysis, BPMN as-is and to-be, a user story map (19 stories) and 22 requirements with MoSCoW priority, traced from each need to code and tests.

SOLUTION

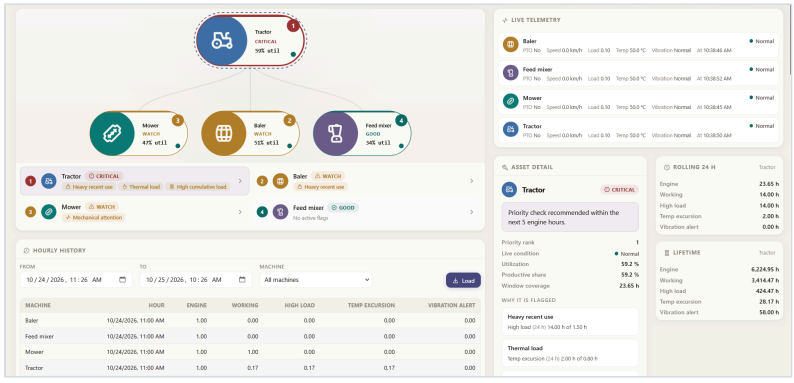
Every morning the farmer opens one dashboard that ranks the four machines and shows status, a service recommendation and the measured reason behind every flag. No hidden scoring, no machine learning.

RESULT

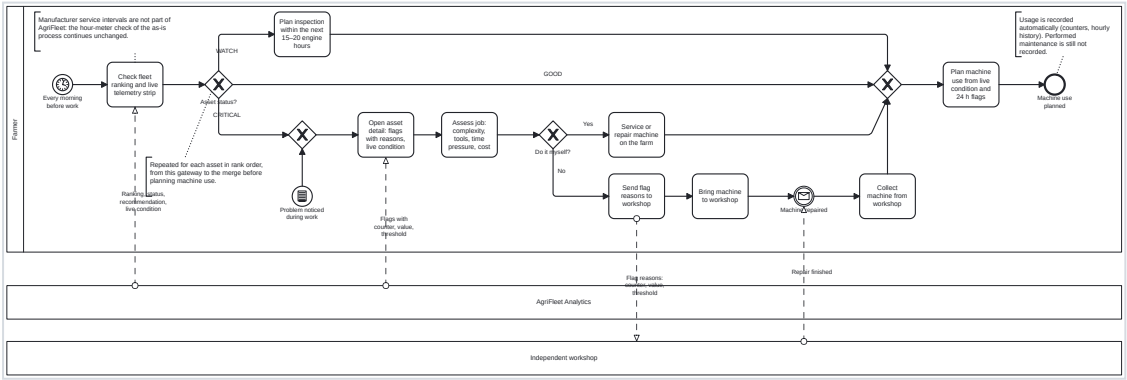
The Must release runs as a distributed system of three Spring Boot modules. Of the 22 requirements, 18 are verified by unit tests and 4 by inspection or demonstration.

Aspect	As-Is	To-Be with AgriFleet
Trigger for a check	Hour meter from memory, or a breakdown	Daily check of the ranked fleet
Order across the fleet	None; each machine judged alone	Priority rank from flags and load
Info for the workshop	Verbal problem description	Counter, value and threshold
Usage record	Nothing is recorded	Automatic hourly history

Unchanged by design: manufacturer service intervals stay a manual hour-meter check, and performed maintenance is still not recorded.



Dashboard: fleet ranked by urgency, live telemetry and the reason behind each flag.



To-Be: each machine is handled by status; the workshop gets flag reasons, not a verbal description.

Stack: Java 25, Spring Boot, RabbitMQ, PostgreSQL, Docker Compose, plain HTML/CSS/JS

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